```
100 'COMBINING RECURSION COEFFICIENTS OF CASCADE AND PARALLEL STAGES
110 '
120 '                                   'INITIALIZE VARIABLES
130 DIM A1[8], B1[8]                    'a and b coefficients for system 1, one of the stages
140 DIM A2[8], B2[8]                    'a and b coefficients for system 2, one of the stages
150 DIM A3[16], B3[16]                  'a and b coefficients for system 3, the combined system
160 '
170                                     'Indicate cascade or parallel combination
180 INPUT "Enter 0 for cascade, 1 for parallel: ", CP%
190 '
200 GOSUB XXXX                          'Mythical subroutine to load: A1[ ], B1[ ], A2[ ], B2[ ]
210 '
220 FOR I% = 0 TO 8                     'Convert the recursion coefficients into transfer functions
230   B2[I%] = -B2[I%]
240   B1[I%] = -B1[I%]
250 NEXT I%
260 B1[0] = 1
270 B2[0] = 1
280 '
290 FOR I% = 0 TO 16                    'Multiply the polynomials by convolving
300   A3[I%] = 0
310   B3[I%] = 0
320   FOR J% = 0 TO 8
330     IF I%-J% < 0  OR I%-J% > 8 THEN GOTO 370
340     IF CP% = 0 THEN A3[I%] = A3[I%] + A1[J%] * A2[I%-J%]
350     IF CP% = 1 THEN A3[I%] = A3[I%] + A1[J%] * B2[I%-J%] + A2[J%] * B1[I%-J%]
360     B3[I%] = B3[I%] + B1[J%] * B2[I%-J%]
370   NEXT J%
380 NEXT I%
390 '
400 FOR I% = 0 TO 16                    'Convert the transfer function into recursion coefficients.
410   B3[I%] = -B3[I%]
420 NEXT I%
430 B3[0] = 0
440 '                                   'The recursion coefficients of the combined system now
450 END                                 'reside in A3[ ] & B3[ ]
```

TABLE 33-1
Combining cascade and parallel stages. This program combines the recursion coefficients of stages in cascade or parallel. The recursive coefficients for the two stages being combined enter the program in the arrays: A1[ ], B1[ ], & A2[ ], B2[ ]. The recursion coefficients that implement the entire system leave the program in the arrays: A3[ ], B3[ ].

This program operates by changing the recursive coefficients from each of the individual stages into transfer functions in the form of Eq. 33-3 (lines 220-270). After combining these transfer functions in the appropriate manner (lines 290-380), the information is moved back to being recursive coefficients (lines 400 to 430).

The heart of this program is how the transfer function polynomials are represented and combined. For example, the numerator of the first stage being combined is: $a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} \cdots$. This polynomial is represented in the program by storing the coefficients: $a_0, a_1, a_2, a_3 \cdots$, in the array: A1[0], A1[1], A1[2], A1[3]$\cdots$. Likewise, the numerator for the second stage is represented by the values stored in: A2[0], A2[1], A2[2], A2[3]$\cdots$, and the numerator for the combined system in: A3[0], A3[1], A3[2], A3[3]$\cdots$. The